Assignment - IA

Design and analysis of Algorithms

Name : A. Vishwajith

Roll no : 241CS106

Section : S7

Course Code : CS253

Scanned with OKEN Scanner

1] Given that,

input size $n = 100$; Time taken for n-input = 0.5 sec,

i] $T(n) = n$; i.e., Linear

$$T(n) \propto n \Rightarrow \frac{100}{0.5} = \frac{1000}{x} \quad (\because \text{Speed is same})$$

$$x = 5 \text{ sec}$$

ii] $T(n) = n^2 \ (\propto n^2)$; Quadratic

$$\frac{(100)^2}{0.5} = \frac{(1000)^2}{x} \Rightarrow x = \frac{0.5 \times 10^6}{10^4} = 50 \text{ sec}$$

iii] Cubic $T(n) = n^3$

$$\frac{(100)^3}{0.5} = \frac{(1000)^3}{x} \Rightarrow x = \frac{0.5 \times 10^9}{10^6} = 500 \text{ sec}$$

2] Given that,

Time for 1 instruction $= 10^{-9}$ sec

no. of sec's in one hour $= 60^2 = 3600$ sec

no. of instructions in one hour $= \frac{3600}{10^{-9}}$ instructions

Processor speed $= 3.6 \times 10^{12}$ instructions

i] Linear Time (n) $\rightarrow T(n) = n$

Steps taken $= n$

Total steps taken should be less than or equal to total instructions possible

$$\boxed{\therefore n \leq 3.6 \times 10^{12}}$$

Scanned with OKEN Scanner

Similarly;

ii] $T(n) = n^2$

$$n^2 \le 3.6 \times 10^{12}$$

$$n \le (3.6)^{1/2} \times 10^6$$

$$\boxed{\therefore n \le 1.89 \times 10^6}$$

iii] $T(n) = n^3$

$$n^3 \le 3.6 \times 10^{12}$$

$$n \le (3.6)^{1/3} \times 10^4$$

$$\boxed{\therefore n \le 1.53 \times 10^4}$$

iv] Exponent of $1.3 \Rightarrow 1.3^n$

$$1.3^n \le 3.6 \times 10^{12}$$

$$n \le \frac{\log(3.6 \times 10^{12})}{\log 1.3}$$

We know that, $\frac{\log_c a}{\log_c b} = \log_b a$

$$\therefore n \le \log_{1.3} 3.6 \times 10^{12}$$

$$\boxed{n \le 110.19}$$

3] Given that,

Running time for insertion sort $\Rightarrow T(n) = 8n^2$

Running time for counting sort $\Rightarrow T(n) = 64n$

Let k be an integer such that $\forall n \le k \Rightarrow \underset{\text{insertion sort}}{T(k)} \le \underset{\text{counting sort}}{T(k)}$

i.e., $8n^2 \le 64n \quad \forall n \le k$

$$8k^2 \le 64k$$

$$\boxed{k \le 8}$$

Scanned with OKEN Scanner

4] Given that,

$T(n) = 5\times10^{-9}\times n^3$ sec

One week time $= 7\times24\times3600$ sec

$= 6.048\times10^5$ sec

Also, Time taken by algorithm $\leq$ Total available time

$$5\times10^{-9}\times n^3 \leq 6.048\times10^5$$

$$n^3 \leq \frac{6.048}{5}\times10^{14}$$

$$n^3 \leq 1.2096\times10^{14}$$

$$n^3 \leq 120.96\times10^{12}$$

$$n \leq (120.96)^{1/3}\times10^4$$

$$n \leq 4.94\times10^4$$

5] Given that,

$f(n) = n + n\log n^2 = n + 2n\log n$

By O-notation $\Rightarrow$ Upper bound / Worst case scenario in Asymptotic analysis

We know that,

$\log n > 1 \quad \forall n > 2$

$\Rightarrow n\log n > n \quad \forall n > 2$

$\therefore n + 2n\log n < 3n\log n \quad \forall n > 2$

We know that,

if $f(n) \leq c.g(n) \quad \forall n \geq n_0 \;\&\; c > 0$ then $f(n) = O(g(n))$

then $\boxed{f(n) = O(n\log n)}$ $\forall n \geq n_0$ $(n_0 = 2)$, $(c = 3 > 0)$

Scanned with OKEN Scanner

6]

Given that,

$f(n) = n\log^2 n$ & $g(n) = n^{0.3}\log^{36} n$

$$\Rightarrow \frac{f(n)}{g(n)} = \frac{n\log^2 n}{n^{0.3}\log^{36} n} = n^{0.7}\log^{-34} n$$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \frac{n^{0.7}}{\log^{34} n} = \infty$$

This implies that $f(n)$ grows much faster than $g(n)$ for n (as it increases).

So, $f(n)$ is faster.

7]

To prove that

$f(n) = O(g(n))$ if and only if $g(n) = \Omega(f(n))$

By definition of $f(n) = O(g(n))$

$\Rightarrow f(n) \le c \cdot g(n) \quad \forall n \ge n_0$ & $c > 0$ —— ①

to prove $g(n) = \Omega(f(n))$ by definition.

we have to prove that

$g(n) \ge d\, f(n) \quad \forall n \ge n_0'$ & $d > 0$

from ①;

$g(n) \ge \frac{1}{c} f(n)$ (possible since $c > 0$ also $1/c > 0$)

$g(n) \ge d\, f(n)$ (Let $1/c = d$)

$\forall n \ge n_0$

So by definition

i.e., $n_0' = n_0$

$\therefore g(n) = \Omega(f(n))$ //

Scanned with OKEN Scanner

8] Let us assume that,

$$f(n) + g(n) = \theta(\min(f(n), g(n))) \quad \text{—— (1)}$$

Then it must be true for $f(n) = n^2$, $g(n) = n$

$\forall n \geq 1 \quad \min(n^2, n) = n \quad (\text{as } n^2 > n)$

then by (1);

$$n^2 + n = \theta(n)$$

This isn't true,

$\therefore$ By contradiction $f(n) + g(n) = \theta(\min(f(n), g(n)))$ is false.

9] Given that,

$$g(n) = o(f(n))$$

To prove $f(n) + g(n) = \theta f(n)$

By definition of $g(n) = o(f(n)) \Rightarrow \lim_{n \to \infty} \frac{g(n)}{f(n)} = 0$ —— (1)

$g(n) \leq c\, f(n) \quad \forall n \geq n_0 \;\&\; c > 0$

$\frac{g(n)}{f(n)} \leq c$ —— (1)

$f(n) + g(n) = f(n)\left(1 + \frac{g(n)}{f(n)}\right)$ from (1)

$\therefore f(n) + g(n) \leq f(n)(1 + c)$

$f(n) + g(n) = f(n)\left(1 + \frac{g(n)}{f(n)}\right)$

$\lim_{n \to \infty} f(n) + g(n) = \lim_{n \to \infty} f(n)\left(1 + \frac{g(n)}{f(n)}\right) = f(n)\left(1 + \lim_{n \to \infty} \frac{g(n)}{f(n)}\right) = f(n)$

$\therefore f(n) + g(n) = \theta(f(n))$

Scanned with OKEN Scanner

10] To prove that,

$a^n = \omega(n^k) \quad \forall\, a > 1$ & $a \in$ Real numbers & $k \geq 1 \quad k \in Z$

$$\underset{n\to\infty}{Lt} \frac{a^n}{n^k} = \frac{a^n \log a}{k \cdot n^{k-1}} \quad \text{(By L-Hospital rule)}$$

By doing it k times.

$$\underset{n\to\infty}{Lt} \frac{a^n \underbrace{\log a \cdot \log a \cdots}_{k \text{ times}}}{k!}$$

$$\frac{(\log a)^k}{k!} \underset{n\to\infty}{Lt}\, a^n \quad (\text{as } a > \phi)$$

$$\underset{n\to\infty}{Lt} \frac{a^n}{n^k} = \infty \Rightarrow a^n = \omega(n^k)$$

11] $\log(n!) = O(n \log n) \rightarrow$ To be proved.

$\log(n!) = \log(n \times n-1 \times \cdots \times 1)$.

$\log(n!) = \log n + \log n-1 + \log n-2 + \cdots + \log 1$

$\forall k < n \quad \log k < \log n$ so,

$$\log(n!) \leq \underbrace{\log n + \log n + \cdots + \log n}_{n \text{ times}}$$

$\log(n!) \leq 1 . n \log n$

Let $1 = c$,

$\log(n!) \leq c\, n \log n \quad \forall\, n \geq 1$ & $c = 1$

Also $n! \geq \left(\frac{n}{2}\right)^{n/2}$

By definition,

$\log(n!) = \theta(n \log n)$

Scanned with OKEN Scanner

$\left(\frac{n}{2}\right)^{n/2} \le n!$ because,

$\frac{n}{2}+1 \ge \frac{n}{2}$

$\frac{n}{2}+2 \ge \frac{n}{2}$

$\vdots$

$\Rightarrow \left(\frac{n}{2}+1\right)\left(\frac{n}{2}+2\right)\cdots\left(\frac{n}{2}+\frac{n}{2}\right) \ge \left(\frac{n}{2}\right)^{n/2}$.

$\frac{n!}{\left(\frac{n}{2}\right)!} \ge \left(\frac{n}{2}\right)^{n/2}$

Also $\left(1.2\cdots\cdots\frac{n}{2}\cdot\left(\frac{n}{2}+1\right)\cdots\left(\frac{n}{2}+\frac{n}{2}\right)\right) = n!$

$\therefore n! \ge \left(\frac{n}{2}\right)^{n/2}$; $n! \le n^n$

$\log n! \ge \frac{n}{2}\log\frac{n}{2}$ & $\log n! \le n\log n$

By Definition

$\log(n!) = \theta(n\log n)$.

12] Given that,

$n! = \omega(2^n)$

Let $n! = f(n)$ $2^n = g(n)$

$$\lim_{n\to\infty}\frac{f(n)}{g(n)} = \frac{n!}{2^n} = \infty$$

as $\frac{n!}{2^n} = \frac{n}{2}\cdot\frac{(n-1)}{2}\cdots\cdots\frac{2}{2}\times\frac{1}{2}$

All $\frac{3}{2}, \frac{4}{2}\cdots\cdots\frac{n}{2} > 1$

So $\forall n \ge 2$ $\frac{f(n)}{g(n)} \ge 1 \Rightarrow \lim_{n\to\infty}\frac{f(n)}{g(n)} = \infty$

$\therefore n! = \omega(2^n)$

Scanned with OKEN Scanner

14

[3] Given that,

To prove $1^2+2^2+\ldots\ldots+n^2 = \frac{n(n+1)(2n+1)}{6}$

By mathematical induction;

i] Let base case be $n=1$

for $n=1$

$$1^2 = \frac{1(2)(3)}{6}$$

$$1 = 1$$

LHS = RHS $\Rightarrow$ base case is true.

ii] Let statement be true for $n=k$

then $1^2+2^2+\ldots\ldots+k^2 = \frac{k(k+1)(2k+1)}{6}$ —— ①

iii] To prove statement is true for $n=k+1$

Add $(k+1)^2$ on both sides

$$\sum_{i=1}^{k}(k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$= (k+1)\left(\frac{2k^2+k+6k+6}{6}\right)$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

$$= \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}$$

$\therefore$ Statement is true for $n=k+1$.

By mathematical induction given statement is true for all natural numbers.

Scanned with OKEN Scanner

14] To prove, for all integers $k \geq 1$ & some $\varepsilon > 0$

$\log^k n = o(n^\varepsilon)$.

i] for $k=1$; (Base case) $\lim_{n\to\infty} \frac{\log n}{n^\varepsilon}$; if $\varepsilon = 1$, $\lim_{n\to\infty} \frac{\log n}{n^\varepsilon} = 0 \Rightarrow$ true

$\log n = o(n)$ as $\log n < cn$ for some $c > 0$

so, $\varepsilon = 1$,,

ii] Assuming statement being true for $k = k'$

$\log^{k'} n = o(n^\varepsilon) \Rightarrow \cancel{\log^k n < c \cdot n^\varepsilon}$ $\lim_{n\to\infty} \frac{\log^{k'} n}{n^\varepsilon} = 0$ —— ①

$\lim (\varepsilon = 1) \forall \varepsilon > 0$

iii] To prove statement is true for $k'+1$

from 1,

$$\lim_{n\to\infty} \frac{\log^{k'} n}{n^\varepsilon} = 0 \quad \text{for } \varepsilon = 1$$

(By L'Hospital's Rule)

$$\lim_{n\to\infty} \frac{\log^{k'+1} n}{n^\varepsilon} = \lim_{n\to\infty} \frac{\log^{k'} n \cdot \frac{1}{n} \cdot (k'+1)}{\varepsilon n^{\varepsilon-1}} = \frac{k'+1}{\varepsilon} \lim_{n\to\infty} \frac{\log^{k'} n}{n^\varepsilon} = 0$$

as, from ①

$$\lim_{n\to\infty} \frac{\log^{k'} n}{n^\varepsilon} = 0 \Rightarrow \lim_{n\to\infty} \frac{\log^{k'+1} n}{n^\varepsilon} = 0$$

$\therefore$ By Induction statement is true $\forall k \geq 1$

15] $(n+a)^b = \Theta(n^b) \rightarrow$ To prove $\forall a, b \in \mathbb{R}$ & $b > 0$

$$(n+a)^b = n^b\left(1 + \frac{a}{n}\right)^b$$

$$\lim_{n\to\infty} n^b\left(1 + \frac{a}{n}\right)^b$$

As $n \to \infty$ $\frac{a}{n} \to \infty$; then by expansion.

Scanned with OKEN Scanner

$$\left(1+\frac{a}{n}\right)^b = 1 + \frac{ba}{n} + \frac{b(b-1)a^2}{2!\,n^2} + \cdots\cdots$$

as $n \to \infty$ all terms $\to 0$

$$\lim_{n\to\infty}\left(1+\frac{a}{n}\right)^b \Longrightarrow 1$$

so, $(n+a)^b = n^b \times k$ (for large n values)

$\therefore$ $\exists$ $c_1, c_2 > 0$ & n_0 such that

$$c_1 n^b \le (n+a)^b \le c_2 n^b \quad \forall\, n \ge n_0$$

$$\Rightarrow (n+a)^b = \Theta(n^b)$$

[6] Given that, $f(n) > 0$, $g(n) > 0$

To prove that $\max(f(n), g(n)) = \Theta(f(n) + g(n))$

We know that,

$$f(n) + g(n) \ge \max(f(n), g(n)) \quad \text{—— ①}$$

and $2\max(f(n), g(n)) \ge f(n) + g(n)$ —— ②

from ①, ②

$$\frac{1}{2}(f(n) + g(n)) \le \max(f(n), g(n)) \le f(n) + g(n)$$

$c_1 = 1/2$, $c_2 = 1$

By definition,

$$\max(f(n), g(n)) = \Theta(f(n) + g(n)) \quad //$$

Scanned with OKEN Scanner